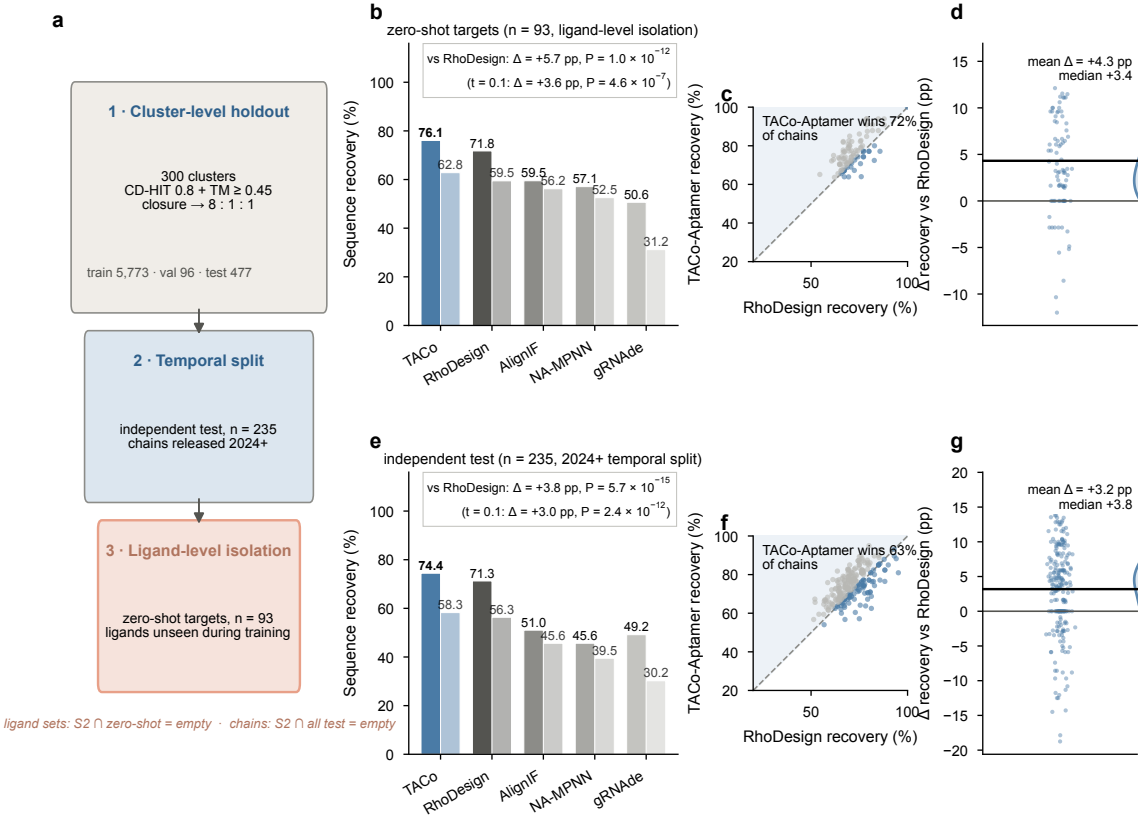

a
1 · Cluster-level holdout
300 clusters
CD-HIT 0.8 + TM ≥ 0.45
closure → 8 : 1 : 1
train 5,773 · val 96 · test 477
2 · Temporal split
independent test, n = 235
chains released 2024+
3 · Ligand-level isolation
zero-shot targets, n = 93
ligands unseen during training
ligand sets: S2 ∩ zero-shot = empty · chains: S2 ∩ all test = empty
b
zero-shot targets (n = 93, ligand-level isolation)
vs RhoDesign: Δ = +5.7 pp, P = 1.0 × 10−12
(t = 0.1: Δ = +3.6 pp, P = 4.6 × 10−7)
Sequence recovery (%)
76.1
62.8
71.8
59.5
59.5
56.2
57.1
52.5
50.6
31.2
TACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
c
TACo-Aptamer recovery (%)
TACo-Aptamer wins 72% of chains
RhoDesign recovery (%)
d
Δ recovery vs RhoDesign (pp)
mean Δ = +4.3 pp
median +3.4
e
independent test (n = 235, 2024+ temporal split)
vs RhoDesign: Δ = +3.8 pp, P = 5.7 × 10−15
(t = 0.1: Δ = +3.0 pp, P = 2.4 × 10−12)
Sequence recovery (%)
74.4
58.3
71.3
56.3
51.0
45.6
45.6
39.5
49.2
30.2
TACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
f
TACo-Aptamer recovery (%)
TACo-Aptamer wins 63% of chains
RhoDesign recovery (%)
g
Δ recovery vs RhoDesign (pp)
mean Δ = +3.2 pp
median +3.8